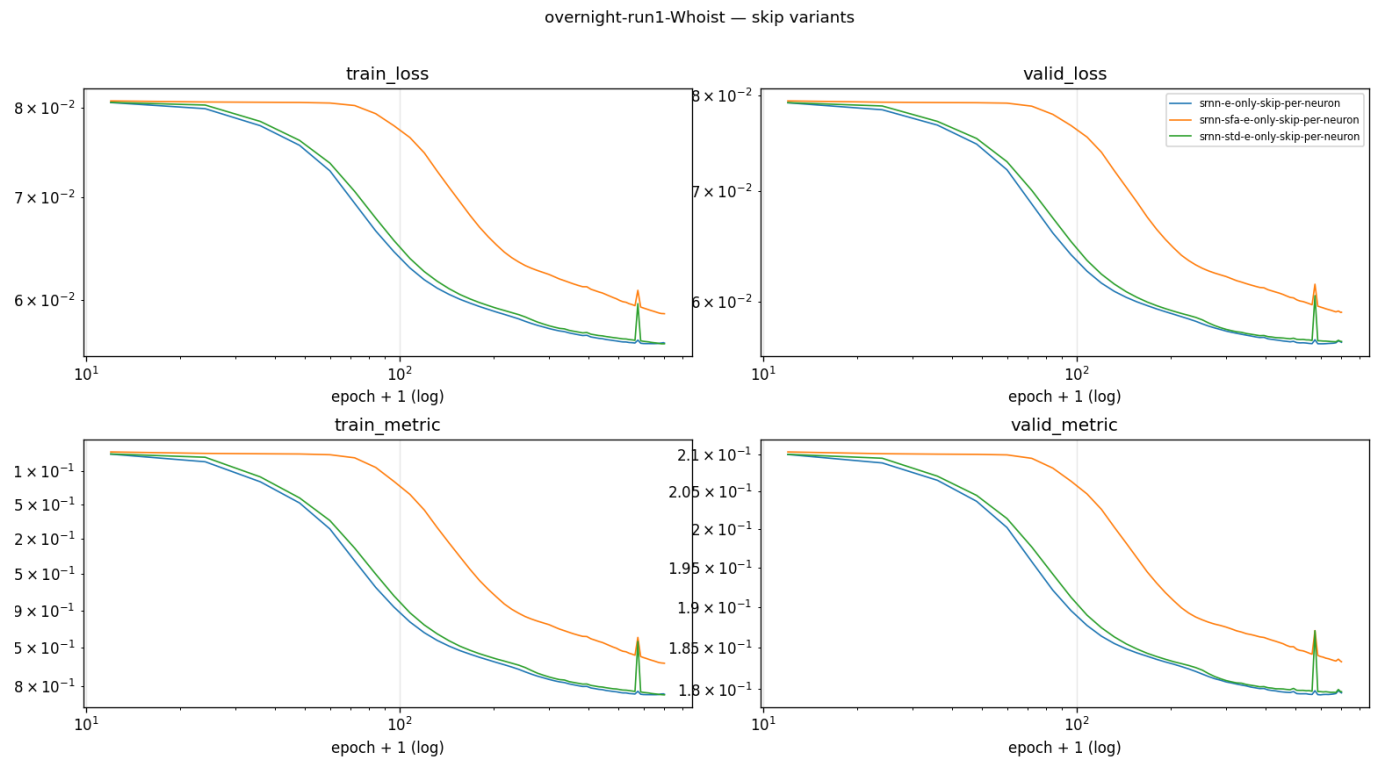


Run: overnight-run1-Whoist

Log-log loss curves



Final test_loss (init - last) [skip]:
 srnn-e-only-skip-per-neuron: 0.07877 → 0.05480
 srnn-sfa-e-only-skip-per-neuron: 0.09420 → 0.05731
 srnn-std-e-only-skip-per-neuron: 0.07907 → 0.05476

Figure 1: skip variants — log-log loss/metric

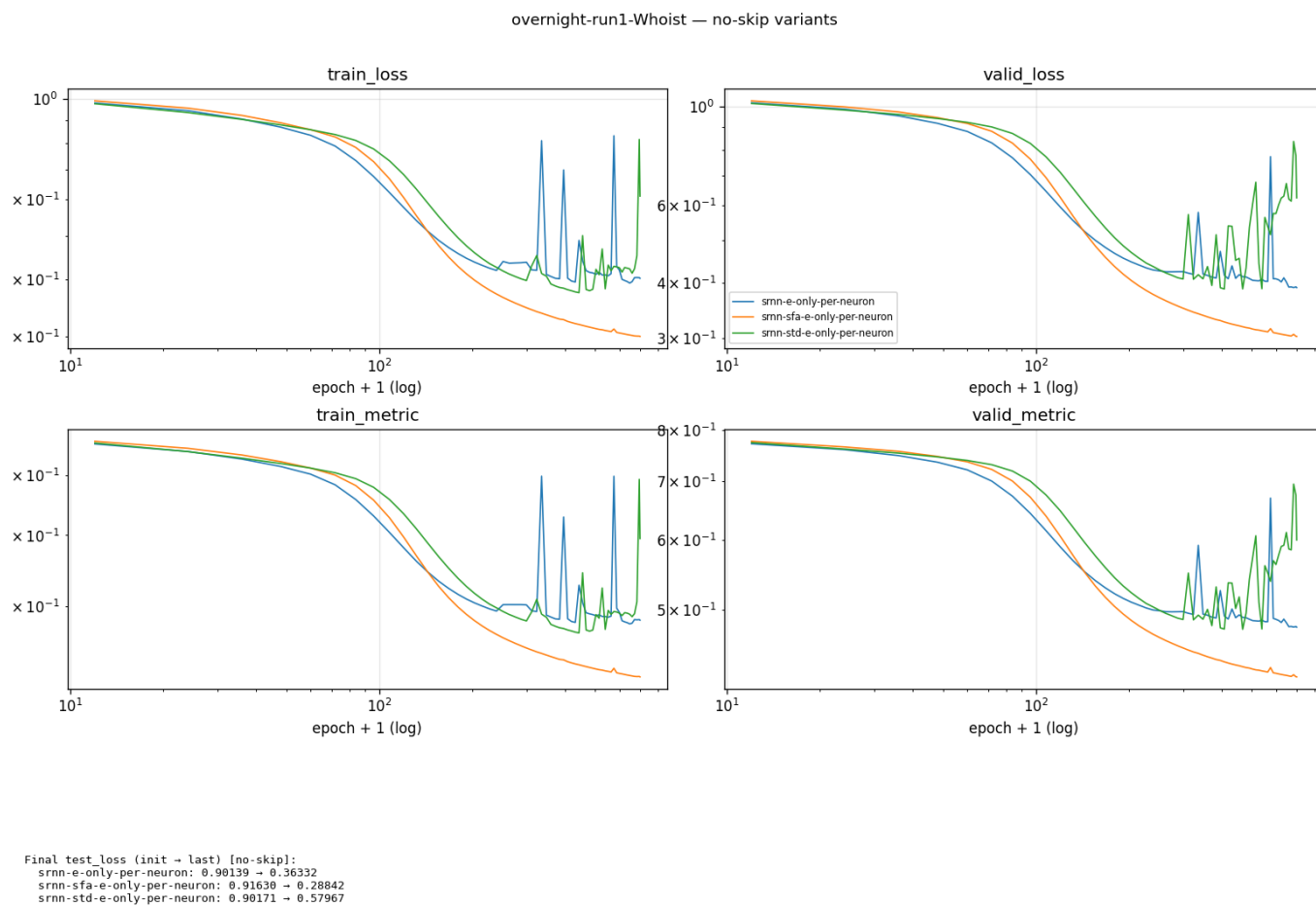


Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

overnight-run1-Whoist — effective time constants (srnn-e-only-per-neuron)

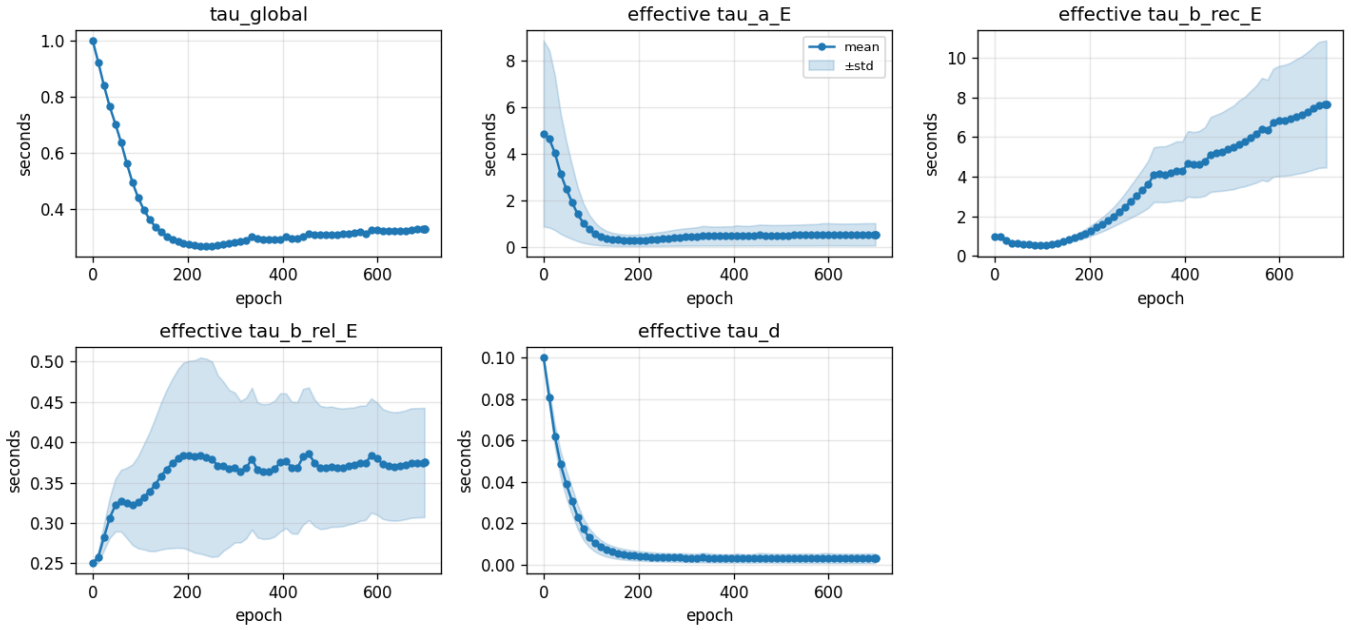
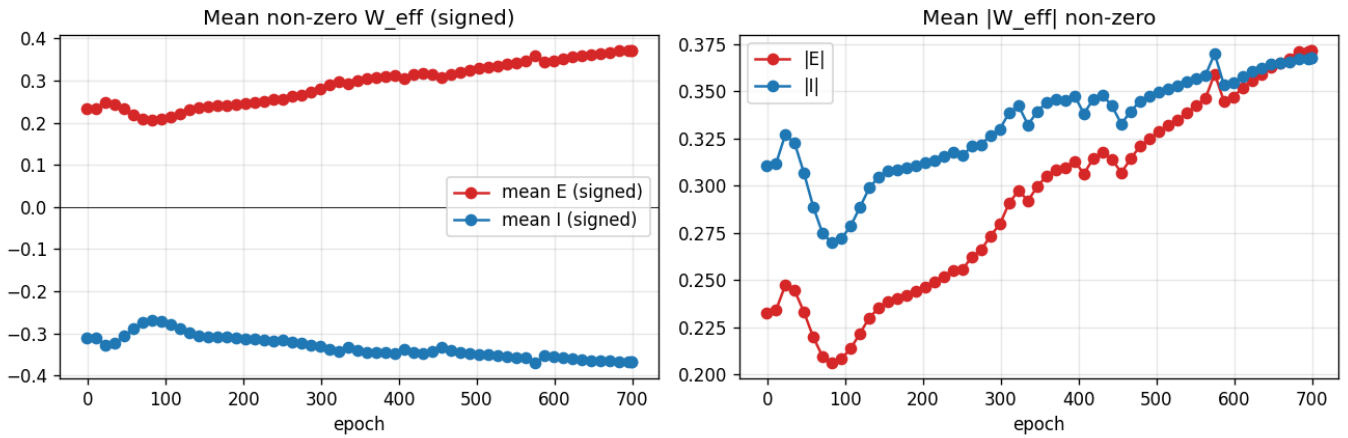


Figure 3: Tau evolution

overnight-run1-Whoist — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# overnight-run1-Whoist - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.32738	-	-	scalar
tau_d (s)	+0.1	+0.0029804	0	0.002336	300
tau_a_E (s)	+4.8777	+0.54953	3.996	0.4781	450
tau_b_rec_E (s)	+1	+7.6743	0	3.204	150
tau_b_rel_E (s)	+0.25	+0.37496	0	0.06777	150
W_eff (E src, signed)	+0.23267	+0.37141	0.07641	0.3083	15048
W_eff (I src, signed)	-0.31062	-0.36737	0.07724	0.1441	14977
W_eff (E src)	+0.23267	+0.37141	0.07641	0.3083	15048
W_eff (I src)	+0.31062	+0.36737	0.07724	0.1441	14977
W_in_eff (input neurons)	-0.0003386	-0.00011137	0.1012	0.06014	6675
c_E (SFA coupling)	+0.05	+0.41231	0	0.5972	450
c_0_E (SFA offset)	+0	-1.5192	0	0.6642	450
a_0 (threshold)	+0.35	-1.3069	0	0.2358	300
readout_weight	-2.8854e-05	-0.0088719	0.06604	1.075	6675
readout_bias	+0	+0.036947	0	0.1891	89
ic.ic (per-variant init)	+0.10386	+0.10386	0.3828	0.3828	900

srnn-e-only-skip-per-neuron

overnight-run1-Whoist — effective time constants (srnn-e-only-skip-per-neuron)

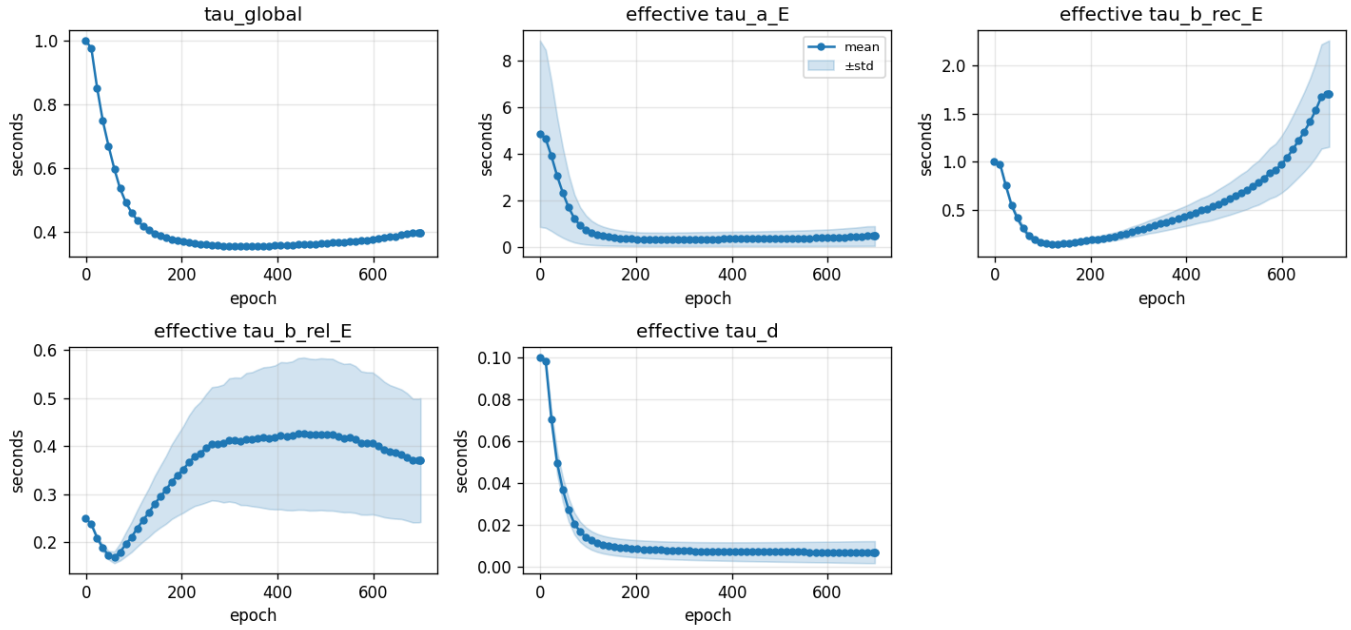
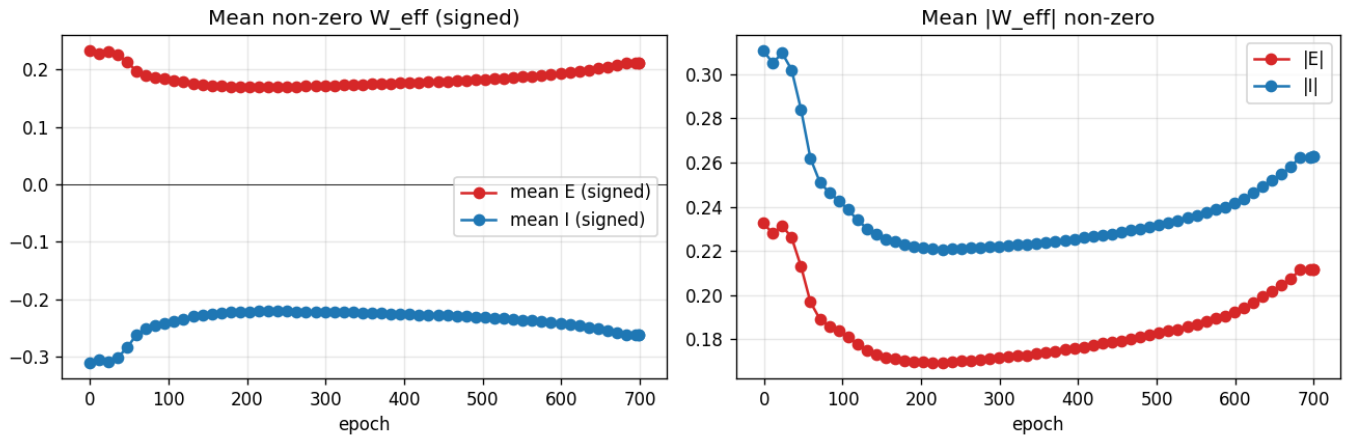


Figure 5: Tau evolution

overnight-run1-Whoist — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

overnight-run1-Whoist - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.39829	-	-	scalar
tau_d (s)	+0.1	+0.0070567	0	0.005277	300
tau_a_E (s)	+4.8777	+0.48719	3.996	0.423	450
tau_b_rec_E (s)	+1	+1.7102	0	0.5535	150
tau_b_rel_E (s)	+0.25	+0.37069	0	0.1286	150
W_eff (E src, signed)	+0.23267	+0.21163	0.07641	0.117	15048
W_eff (I src, signed)	-0.31062	-0.26258	0.07724	0.116	14977
W_eff (E src)	+0.23267	+0.21163	0.07641	0.117	15048
W_eff (I src)	+0.31062	+0.26258	0.07724	0.116	14977
W_in_eff (input neurons)	-0.00084237	-0.000566	0.1008	0.03338	6675
c_E (SFA coupling)	+0.05	+0.1532	0	0.1593	450
c_0_E (SFA offset)	+0	-0.59708	0	0.3098	450
a_0 (threshold)	+0.35	-0.53513	0	0.1452	300
readout_weight	+0.00078219	-0.0020769	0.06654	0.3112	6675
readout_bias	+0	+0.14165	0	0.2093	89
ic.ic (per-variant init)	+0.10327	+0.10327	0.3829	0.3829	900

srnn-sfa-e-only-per-neuron

overnight-run1-Whoist — effective time constants (srnn-sfa-e-only-per-neuron)

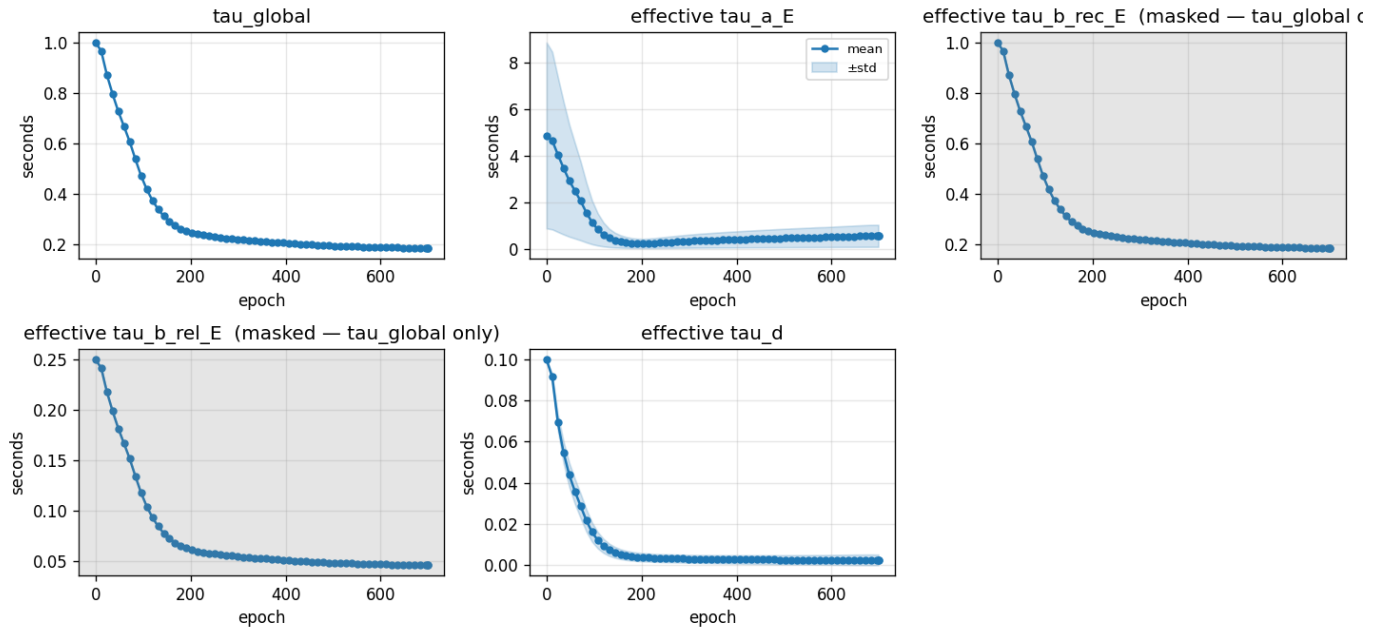
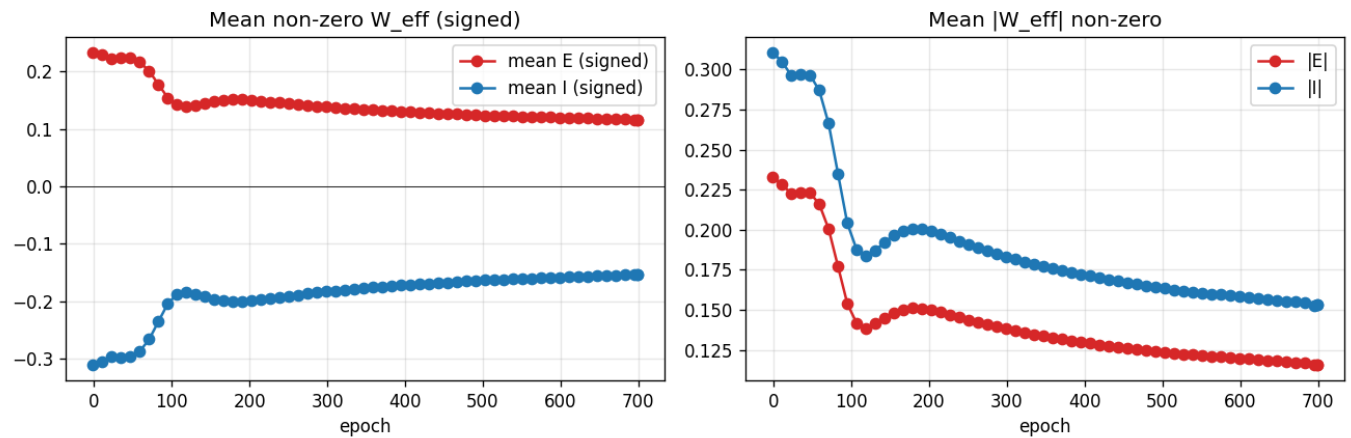


Figure 7: Tau evolution

overnight-run1-Whoist — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# overnight-run1-Whoist - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.18512	-	-	scalar
tau_d (s)	+0.1	+0.0023674	0	0.002576	300
tau_a_E (s)	+4.8777	+0.56125	3.996	0.4747	450
tau_b_rec_E (s)	+1	+0.18512	0	0	150
tau_b_rel_E (s)	+0.25	+0.04628	0	0	150
W_eff (E src, signed)	+0.23267	+0.11601	0.07641	0.0592	15048
W_eff (I src, signed)	-0.31062	-0.15321	0.07724	0.06234	14977
W_eff (E src)	+0.23267	+0.11601	0.07641	0.0592	15048
W_eff (I src)	+0.31062	+0.15321	0.07724	0.06234	14977
W_in_eff (input neurons)	-0.00015382	-0.0010107	0.09934	0.05065	6675
c_E (SFA coupling)	+0.05	+0.41158	0	0.6209	450
c_0_E (SFA offset)	+0	-0.28508	0	0.2741	450
a_0 (threshold)	+0.35	+0.087299	0	0.1382	300
readout_weight	-0.00098972	+0.0081173	0.06678	0.5427	6675
readout_bias	+0	-0.1264	0	0.2967	89
ic.ic (per-variant init)	+0.13078	+0.13078	0.4172	0.4172	900

srnn-sfa-e-only-skip-per-neuron

overnight-run1-Whoist — effective time constants (srnn-sfa-e-only-skip-per-neuron)

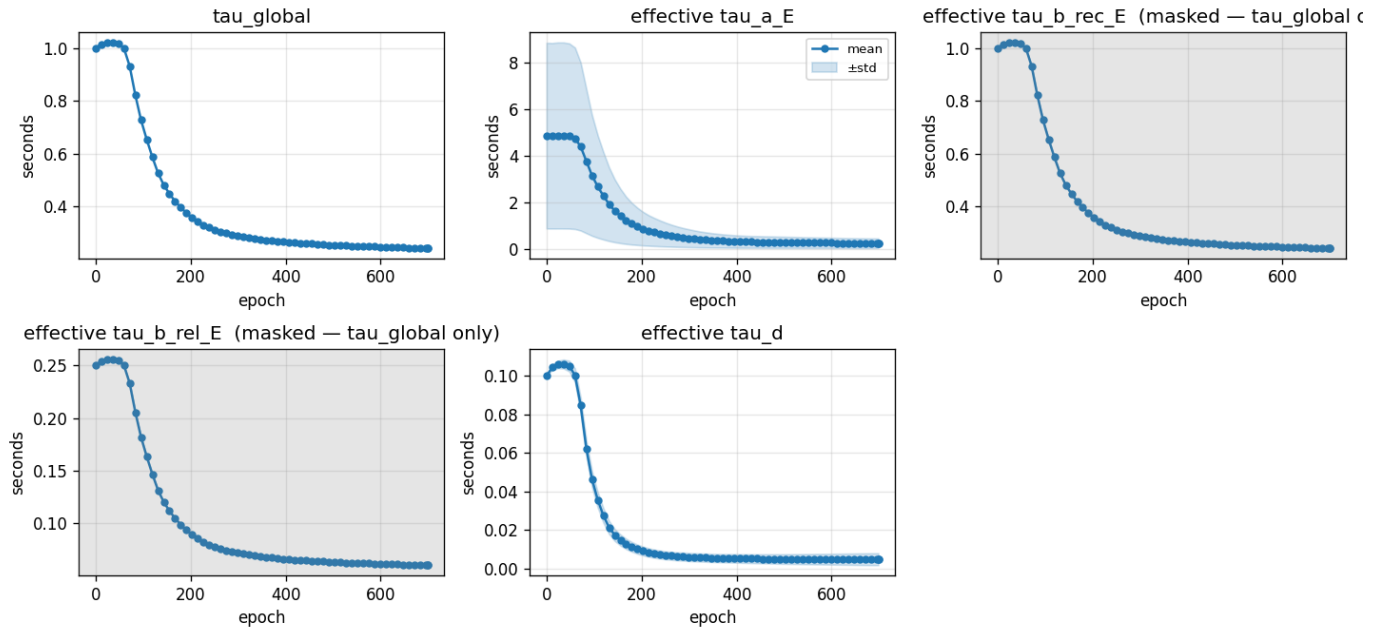
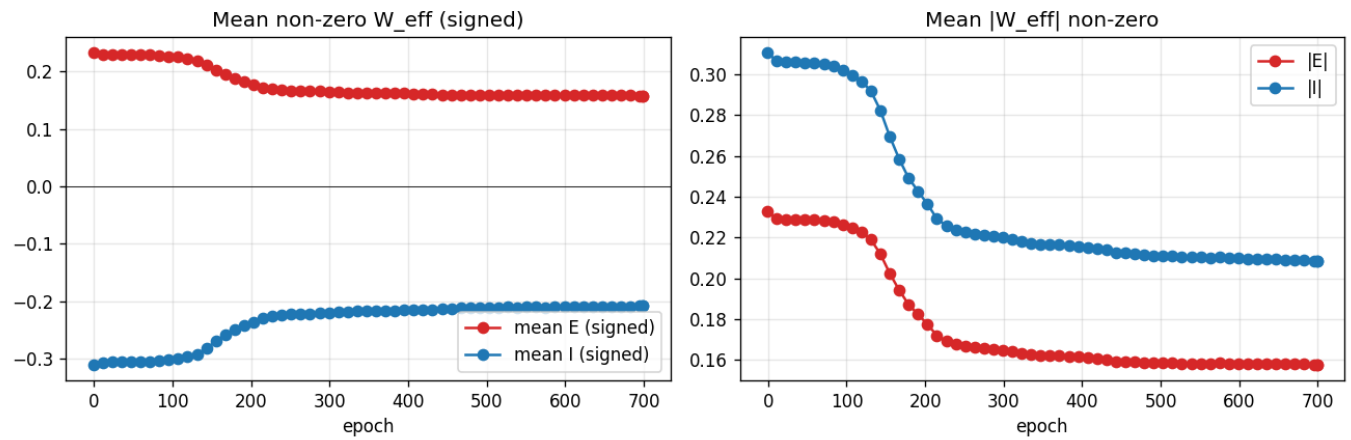


Figure 9: Tau evolution

overnight-run1-Whoist — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# overnight-run1-Whoist - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.2414	-	-	scalar
tau_d (s)	+0.1	+0.0048985	0	0.003099	300
tau_a_E (s)	+4.8777	+0.24446	3.996	0.2074	450
tau_b_rec_E (s)	+1	+0.2414	0	0	150
tau_b_rel_E (s)	+0.25	+0.06035	0	0	150
W_eff (E src, signed)	+0.23267	+0.15765	0.07641	0.0644	15048
W_eff (I src, signed)	-0.31062	-0.20829	0.07724	0.07216	14977
W_eff (E src)	+0.23267	+0.15765	0.07641	0.0644	15048
W_eff (I src)	+0.31062	+0.20829	0.07724	0.07216	14977
W_in_eff (input neurons)	-0.0011621	-0.00010726	0.09962	0.02913	6675
c_E (SFA coupling)	+0.05	+0.072426	0	0.04932	450
c_0_E (SFA offset)	+0	-0.31112	0	0.2576	450
a_0 (threshold)	+0.35	+0.22147	0	0.1239	300
readout_weight	+0.00016801	-0.0088109	0.0673	0.2104	6675
readout_bias	+0	+0.1166	0	0.1961	89
ic.ic (per-variant init)	+0.09231	+0.09231	0.5294	0.5294	900

srnn-std-e-only-per-neuron

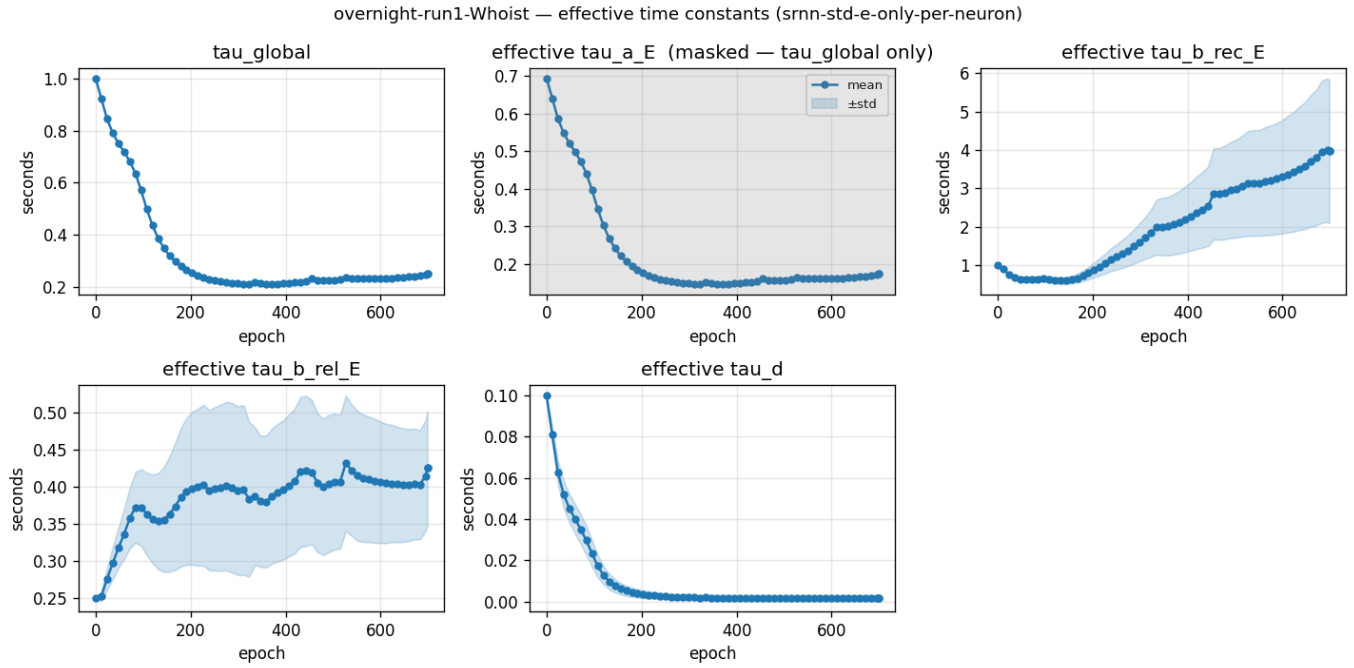
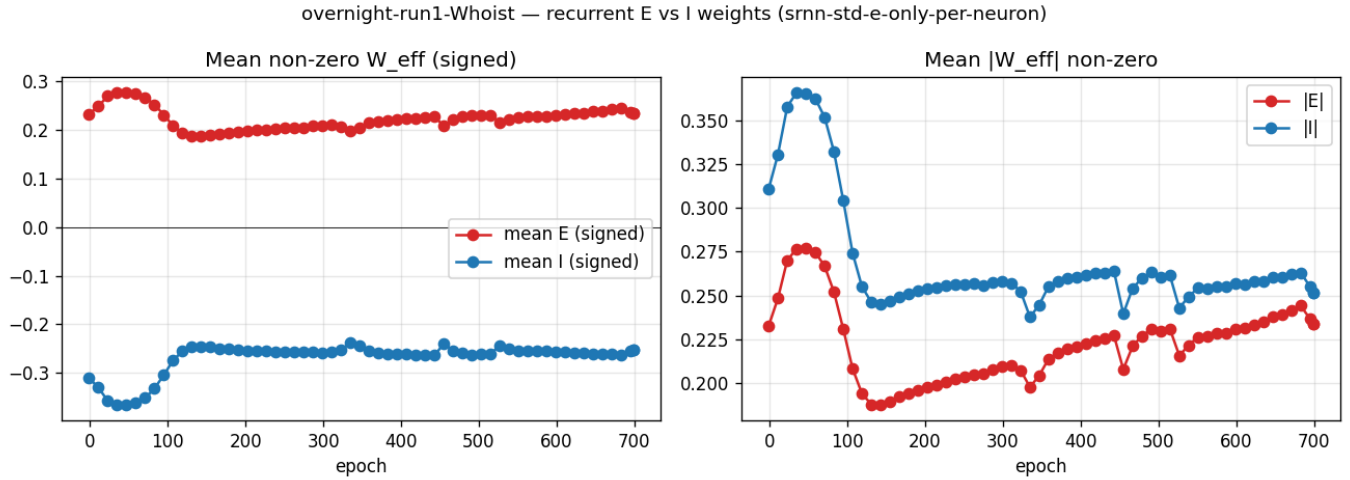


Figure 11: Tau evolution

Figure 12: W_{EI} evolution

srnn-std-e-only-per-neuron — parameter table

```
# overnight-run1-Whoist - variant srnn-std-e-only-per-neuron (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.25008	-	-	scalar
tau_d (s)	+0.1	+0.0017287	0	0.001449	300
tau_a_E (s)	+0.69315	+0.17334	0	0	450
tau_b_rec_E (s)	+1	+3.9623	0	1.862	150
tau_b_rel_E (s)	+0.25	+0.42512	0	0.07687	150
W_eff (E src, signed)	+0.23267	+0.23364	0.07641	0.1586	15048
W_eff (I src, signed)	-0.31062	-0.25145	0.07724	0.09947	14977
W_eff (E src)	+0.23267	+0.23364	0.07641	0.1586	15048
W_eff (I src)	+0.31062	+0.25145	0.07724	0.09947	14977
W_in_eff (input neurons)	+0.00019944	-0.00031341	0.09994	0.03871	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	-1.4128	0	0.2758	300
readout_weight	+0.00074661	-0.0069444	0.06657	0.8627	6675
readout_bias	+0	-0.019712	0	0.1602	89
ic.ic (per-variant init)	+0.090819	+0.090819	0.3835	0.3835	900

srnn-std-e-only-skip-per-neuron

overnight-run1-Whoist — effective time constants (srnn-std-e-only-skip-per-neuron)

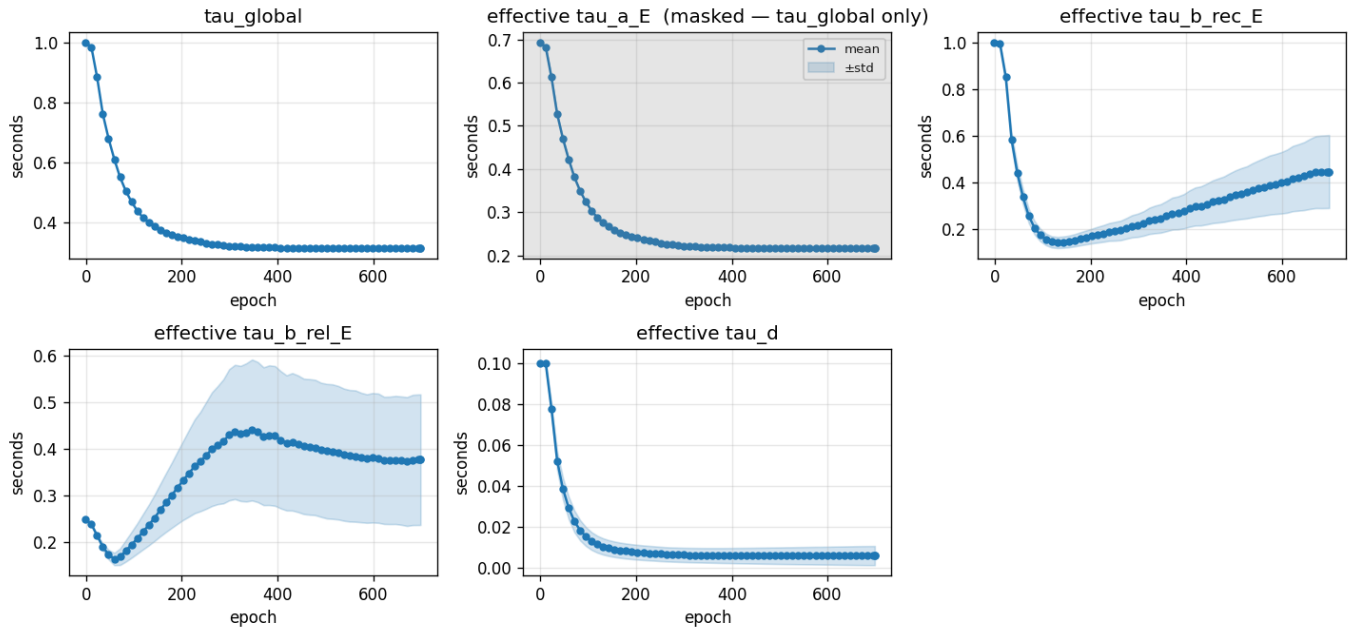
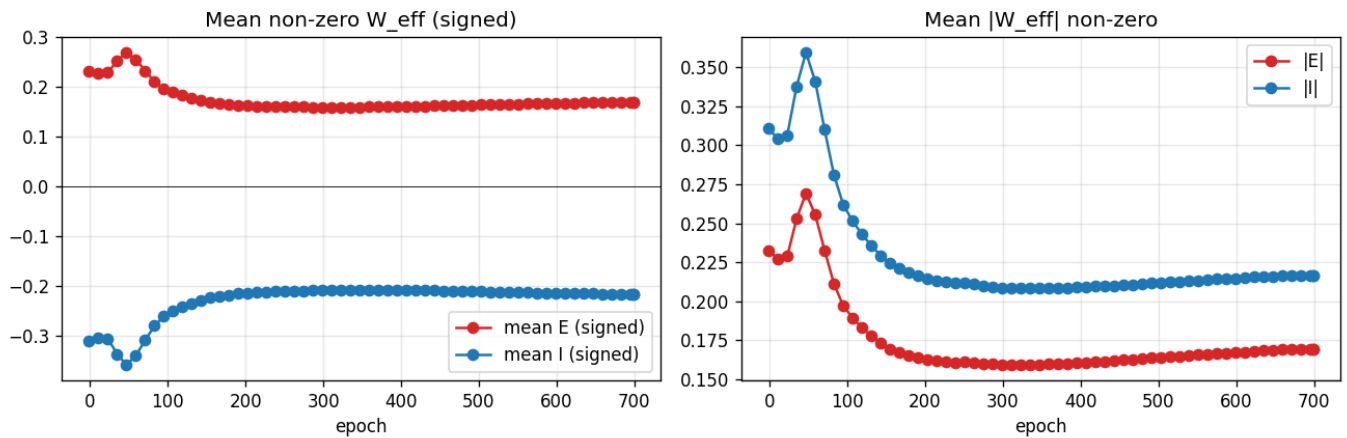


Figure 13: Tau evolution

overnight-run1-Whoist — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

Figure 14: W_{EI} evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# overnight-run1-Whoist - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.31356	-	-	scalar
tau_d (s)	+0.1	+0.0058463	0	0.004701	300
tau_a_E (s)	+0.69315	+0.21734	0	0	450
tau_b_rec_E (s)	+1	+0.44648	0	0.1566	150
tau_b_rel_E (s)	+0.25	+0.37733	0	0.1397	150
W_eff (E src, signed)	+0.23267	+0.16943	0.07641	0.08499	15048
W_eff (I src, signed)	-0.31062	-0.21666	0.07724	0.09173	14977
W_eff (E src)	+0.23267	+0.16943	0.07641	0.08499	15048
W_eff (I src)	+0.31062	+0.21666	0.07724	0.09173	14977
W_in_eff (input neurons)	+0.00036119	+8.7197e-05	0.09956	0.02451	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	-0.33933	0	0.1586	300
readout_weight	+0.00055372	-0.0098317	0.06572	0.2858	6675
readout_bias	+0	-0.0034242	0	0.1832	89
ic.ic (per-variant init)	+0.083543	+0.083543	0.384	0.384	900